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GATE ECE · GENERAL APTITUDE · CHAPTER-WISE PYQS

Numerical Ability

GATE ECE General Aptitude · Previous Year Questions with Answer Key

101	22	2010 – 2026	English
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Marks 1	43 questions
Marks 2	58 questions
Answer key	at the end

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Marks 1 43 questions

Q1. GATE ECE 2026

Open solution

Real numbers y, p, and n (all greater than 1) satisfy

$\left(\log_{p^{1/n}} y\right)\left(\log_{y^{1/n}} p\right) = 16$

where the logarithms are taken to the bases $p^{1/n}$ and $y^{1/n}$ The value of n is ____

(A) 2 (B) 4 (C) 8 (D) 16

Q2. GATE ECE 2025

Open solution

The 12 musical notes are given as $C, C^{\#}, D, D^{\#}, E, F, F^{\#}, G, G^{\#}, A, A^{\#}, B$. Frequency of each note is $\sqrt[12]{2}$ times the frequency of the previous note. If the frequency of the note C is 130.8 Hz , then the ratio of frequencies of notes $F^{\#}$ and C is:

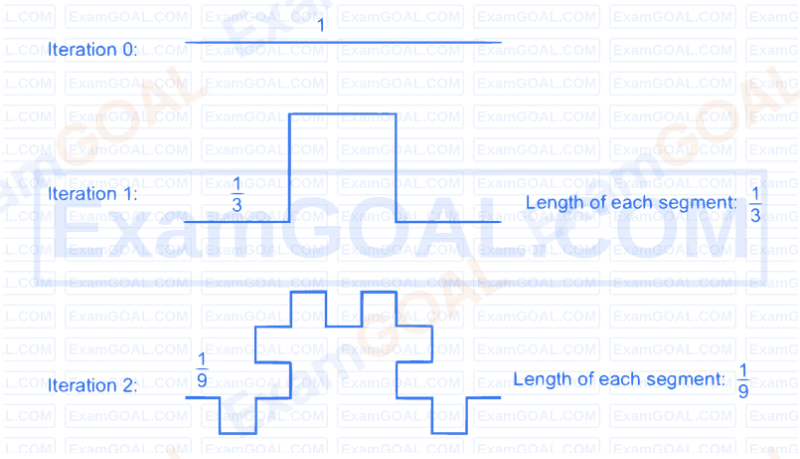
(A) $\sqrt[6]{2}$ (B) $\sqrt{2}$ (C) $\sqrt[4]{2}$ (D) 2

Q3. GATE ECE 2025

Open solution

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration n ' is:

Note: The figures shown are representative.



(A) $\left(\frac{5}{3}\right)^{\frac{n}{2}}$ (B) $\left(\frac{5}{3}\right)^{n}$

(C) $\left(\frac{5}{3}\right)^{2n}$ (D) $\left(\frac{5}{3}\right)^{n(2n-1)}$

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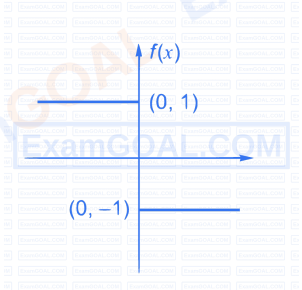
Q4. GATE ECE 2025

Which one of the following plots represents $f(x) = -\frac{|x|}{x}$, where x is a non-zero real number?

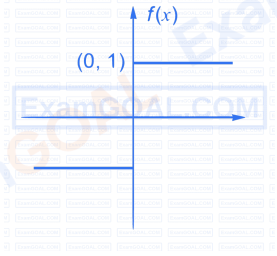
Note: The figures shown are representative.

Open solution

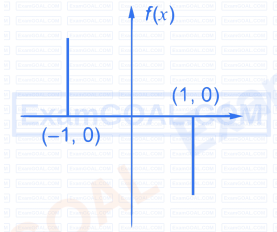
(A)



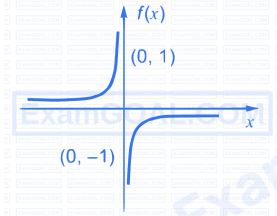
(B)



(C)



(D)



Q5. GATE ECE 2024

Five years ago, the ratio of Aman's age to his father's age was 1:4, and five years from now, the ratio will be 2:5. What was his father's age when Aman was born?

(A) 28 years (B) 30 years (C) 35 years (D) 32 years

Open solution

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Q6. GATE ECE 2024

For a real number $x > 1$,

$\frac{1}{\log_2 x} + \frac{1}{\log_3 x} + \frac{1}{\log_4 x} = 1$

The value of x is

(A) 4 (B) 12 (C) 24 (D) 36

Open solution

Q7. GATE ECE 2024

The greatest prime factor of $(3^{199} - 3^{196})$ is

(A) 13 (B) 17 (C) 3 (D) 11

Open solution

Q8. GATE ECE 2023

What is the smallest number with distinct digits whose digits add up to 45?

(A) 123555789 (B) 123457869 (C) 123456789 (D) 99999

Open solution

Q9. GATE ECE 2023

In a class of 100 students,

(i) there are 30 students who neither like romantic movies nor comedy movies,

(ii) the number of students who like romantic movies is twice the number of students who like comedy movies, and

(iii) the number of students who like both romantic movies and comedy movies is 20.

How many students in the class like romantic movies?

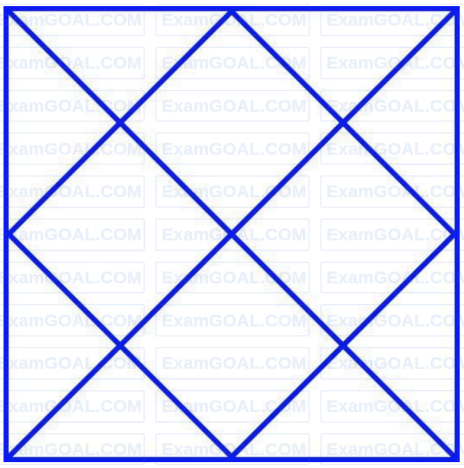
(A) 40 (B) 20 (C) 60 (D) 30

Open solution

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Q10. GATE ECE 2023

How many rectangles are present in the given figure?



Open solution

(A) 8 (B) 9 (C) 10 (D) 12

Q11. GATE ECE 2022

A sum of money is to be distributed among P, Q, R, and S in the proportion 5 : 2 : 4 : 3, respectively.

If R gets Rs. 1000 more than S, what is the share of Q (in Rs.) ?

Open solution

(A) 500 (B) 1000 (C) 1500 (D) 2000

Q12. GATE ECE 2022

A trapezium has vertices marked as P, Q, R and S (in that order anticlockwise). The side PQ is parallel to side SR.

Further, it is given that, PQ = 11 cm, QR = 4 cm, RS = 6 cm and SP = 3 cm. What is the shortest distance between PQ and SR (in cm) ?

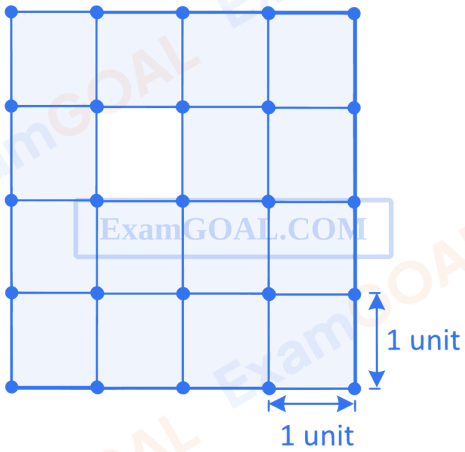
Open solution

(A) 1.80 (B) 2.40 (C) 4.20 (D) 5.76

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Q13. GATE ECE 2022

The figure shows a grid formed by a collection of unit squares. The unshaded unit square in the grid represents a hole.



What is the maximum number of squares without a "hole in the interior" that can be formed within the 4×4 grid using the unit squares as building blocks?

(A) 15 (B) 20 (C) 21 (D) 26

Open solution

Q14. GATE ECE 2022

Four points P(0, 1), Q(0, -3), R(-2, -1), and S(2, -1) represent the vertices of a quadrilateral. What is the area enclosed by the quadrilateral?

(A) 4 (B) $4\sqrt{2}$ (C) 8 (D) $8\sqrt{2}$

Open solution

Q15. GATE ECE 2021

p and q are positive integers and $\frac{p}{q} + \frac{q}{p} = 3$, then $\frac{p^2}{q^2} + \frac{q^2}{p^2}$ =

(A) 11 (B) 3 (C) 9 (D) 7

Open solution

Q16. GATE ECE 2021

The current population of a city is 11,02,500. If it has been increasing at the rate of 5% per annum, what was its population 2 yeas ago?

(A) 9,92,500 (B) 9,95,006 (C) 12,51,506 (D) 10,00,000

Open solution

Q17. GATE ECE 2020

A superadditive function $f(\cdot)$ satisfies the following property

$f(x_1 + x_2) \geq f(x_1) + f(x_2)$

Which of the following functions is a superadditive function for $x > 1$?

(A) $\sqrt{x}$ (B) e^x (C) $\frac{1}{x}$ (D) e^{-x}

Open solution

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Q18. GATE ECE 2019

Five different books (P, Q, R, S, T) are to be arranged on a shelf. The books R and S are to be arranged first and second respectively from right side of the shelf. The number of different orders in which P, Q and T may be arranged is _____.

Open solution

(A) 6 (B) 2 (C) 12 (D) 120

Q19. GATE ECE 2019

It would take one machine 4 hours to complete a production order and another machine 2 hours to complete the same order. If both machine work simultaneously at their respective constant rates, the time taken to complete the same order is _____ hours.

Open solution

(A) $\frac{4}{3}$ (B) $\frac{3}{4}$ (C) $\frac{2}{3}$ (D) $\frac{7}{3}$

Q20. GATE ECE 2018

A 1.5 m tall person is standing at a distance of 3 m from a lamp post. The light from the lamp at the top of the post casts her shadow. The length of the shadow is twice her height. What is the height of the lamp post in meters?

Open solution

(A) 1.5 (B) 3 (C) 4.5 (D) 6

Q21. GATE ECE 2018

If the number 715 ■ 423 is divisible by 3 (■ denotes the missing digit in the thousands place), then the smallest whole number in the place of ■ is _____.

Open solution

(A) 2 (B) 0 (C) 6 (D) 5

Q22. GATE ECE 2018

What is the value of $1+\frac{1}{4}+\frac{1}{16}+\frac{1}{64}+\frac{1}{256}+\ldots\ldots$?

Open solution

(A) $\frac{7}{4}$ (B) $\frac{3}{2}$ (C) $\frac{4}{3}$ (D) 2

Q23. GATE ECE 2017 Set 2

A rule states that in order to drink beer one must be over 18 years old. In a bar, there are 4 people. P is 16 years old, Q is 25 years old, R is drinking milkshake and S is drinking beer. What must be checked to ensure that the rule is being followed?

Open solution

(A) Only P's drink (B) Only P's drink and S's age

(C) Only S's age (D) Only P's drink, Q's drink and S's age

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Q24. GATE ECE 2017 Set 2

Fatima starts from point P, goes North for 3 km, and then East for 4 km to reach point Q. She then turns to face point P and goes 15 km in that direction. She then goes North for 6 km. How far is she from point P, and in which direction should she go to reach point P?

Open solution

(A) 8 km, East (B) 12 km, North (C) 6 km, East (D) 10 km, North

Q25. GATE ECE 2017 Set 2

500 students are taking one or more courses out of Chemistry, Physics, and Mathematics. Registration records indicate course enrolment as follows: Chemistry (329), Physics (186), Mathematics (295), Chemistry and Physics (83), Chemistry and Mathematics (217), and Physics and Mathematics (63). How many students are taking all 3 subjects?

Open solution

(A) 37 (B) 43 (C) 47 (D) 53

Q26. GATE ECE 2017 Set 1

In the summer, water consumption is known to decrease overall by 25%. A water Board official states that in the summer household consumption decreases by 20%, while other consumption increases by 70%.
Which of the following statements is correct?

Open solution

(A) The ratio of household to other consumption is 8/17

(B) The ratio of household to other consumption is 1/17

(C) The ratio of household to other consumption is 17/8

(D) There are errors in the official's statement

Q27. GATE ECE 2017 Set 1

40% of deaths on city roads may be attributed to drunken driving. The number of degrees needed to represent this as a slice of a pie chart is

Open solution

(A) 120 (B) 144 (C) 160 (D) 212

Q28. GATE ECE 2017 Set 1

Some tables are shelves. Some shelves are chairs. All chairs are benches. Which of the following conclusions can be deduced from the preceding sentences?
(i) At least one bench is a table
(ii) At least one shelf is a bench
(iii) At least one chair is a table
(iv) All benches are chairs

Open solution

(A) Only (i) (B) Only (ii)

(C) Only (ii) and (iii) (D) Only (iv)

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Q29. GATE ECE 2016 Set 2

Given $(9 \text{ inches})^{1/2} = (0.25 \text{ yards})^{1/2}$, which one of the following statements is TRUE?

(A) 3 inches = 0.5 yards
(B) 9 inches = 1.5 yards
(C) 9 inches = 0.25 yards
(D) 81 inches = 0.0625 yards

Open solution

Q30. GATE ECE 2016 Set 2

S, M, E and F are working in shifts in a team to finish a project. M works with twice the efficiency of others but for half as many days as E worked. S and M have 6 hour shifts in a day, whereas E and F have 12 hours shifts. What is the ratio of contribution of M to contribution of E in the project?

(A) 1:1 (B) 1:2 (C) 1:4 (D) 2:1

Open solution

Q31. GATE ECE 2016 Set 1

In a huge pile of apples and oranges, both ripe and unripe mixed together, 15% are unripe fruits. Of the unripe fruits, 45% are apples. Of the ripe ones, 66% are oranges. If the pile contains a total of 5692000 fruits, how many of them are apples?

(A) 2029198 (B) 2467482 (C) 2789080 (D) 3577422

Open solution

Q32. GATE ECE 2016 Set 1

Michael lives 10 km away from where I live. Ahmed lives 5 km away and Susan lives 7 km away from where I live. Arun is farther away than Ahmed but closer than Susan from where I live. From the information provided here, what is one possible distance (in km) at which I live from Arun's place?

(A) 3.00 (B) 4.99 (C) 6.02 (D) 7.01

Open solution

Q33. GATE ECE 2016 Set 3

The number that least fits this set: (324, 441, 97 and 64) is _____.

(A) 324 (B) 441 (C) 97 (D) 64

Open solution

Q34. GATE ECE 2016 Set 3

It takes 10 s and 15 s, respectively, for two trains traveling at different constant speeds to completely pass a telegraph post. The length of the first train is 120 m and that of the second train is 150 m. The magnitude of the difference in the speeds of the two trains (in m/s) is _____.

(A) 2.0 (B) 10.0 (C) 12.0 (D) 22.0

Open solution

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Q35. GATE ECE 2015 Set 3

If x>y>1, which of the following must be true?

i. $\ln(x) > \ln(y)$ ii. $e^x > e^y$
iii. $y^x > x^y$ iv. $\cos(x) > \cos(y)$

(A) (i) and (ii) (B) (i) and (iii) (C) (iii) and (iv) (D) (ii) and (iv)

Open solution

Q36. GATE ECE 2015 Set 3

Find the missing sequence in the letter series below:
A, CD, GHI, ? , UVWXY

(A) LMN (B) MNO (C) MNOP (D) NOPQ

Open solution

Q37. GATE ECE 2015 Set 2

Ram and Ramesh appeared in an interview for two vacancies in the same department. The probability of Ram's selection is 1/6 and that of Ramesh is 1/8. What is the probability that only one of them will be selected?

(A) 47/48 (B) 1/4 (C) 13/48 (D) 35/48

Open solution

Q38. GATE ECE 2015 Set 1

If $\log_x\left(\frac{5}{7}\right) = \frac{-1}{3}$, then the value of x is

(A) 343/125 (B) 125/343 (C) -25/49 (D) -49/25

Open solution

Q39. GATE ECE 2015 Set 1

Operators □,◊,→ are defined by : a □ b = $\frac{a-b}{a+b}$; a ◊ b = $\frac{a+b}{a-b}$; a → b = ab. Find the value of (66 □ 6) → (66 ◊ 6).

(A) -2 (B) -1 (C) 1 (D) 2

Open solution

Q40. GATE ECE 2013

In the summer of 2012, in New Delhi, the mean temperature of Monday to Wednesday was 41°C and of Tuesday to Thursday was 43°C. If the temperature on Thursday was 15% higher than that of Monday, then the temperature in °C on Thursday was

(A) 40 (B) 43 (C) 46 (D) 49

Open solution

Q41. GATE ECE 2012

If $(1.001)^{1259} = 3.52$ and $(1.001)^{2062} = 7.85$, then $(1.001)^{3321}$ =

(A) 2.23 (B) 4.33 (C) 11.37 (D) 27.64

Open solution

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Q42. GATE ECE 2011

There are two candidates P and Q in an election. During the campaign, 40% of the voters promised to vote for P, and rest for Q. However, on the day of election 15% of the voters went back on their promise to vote for P and instead voted for Q. 25% of the voters went back on their promise to vote for Q and instead voted for P. Suppose, P lost by 2 votes, then what was the total number of voters?

Open solution

(A) 100 (B) 10 (C) 90 (D) 95

Q43. GATE ECE 2010

25 persons are in a room. 15 of them play hockey, 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is:

Open solution

(A) 2 (B) 17 (C) 13 (D) 3

Marks 2 58 questions

Q44. GATE ECE 2026

Let P_k represent the perimeter of a square with sides of length k. The value of the expression $P_1 + P_2 + P_3 + \ldots + P_{10}$ is ____ .

Open solution

(A) 55 (B) 110 (C) 220 (D) 440

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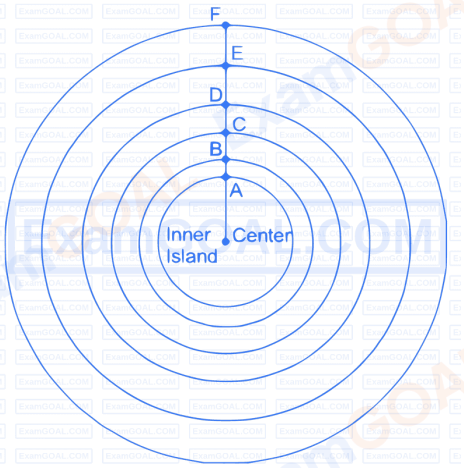
Q45. GATE ECE 2026

Open solution

The city of Atlantis was crafted by the God of the seas, Poseidon. It was made of alternating concentric circular rings of land (shaded) and water (not shaded) as represented in the figure (not to scale). The radius of Inner Island was 2.5 stades (a unit of length used in ancient Greece). The water surrounding Inner Island was one stade wide (length AB). This was surrounded by two pairs of alternating rings of land and water. The first pair of land and water was two stades wide each (lengths BC and CD), and the outer pair is three stades wide each (lengths DE and EF).

The ratio of the surface area of the land to that of the water in the city of Atlantis is

____ (round off to two decimal places).



(A) 0.45 (B) 0.60 (C) 0.75 (D) 0.90

Q46. GATE ECE 2025

Open solution

A stick of length one meter is broken at two locations at distances of b_1 and b_2 from the origin (0), as shown in the figure. Note that $0 < b_1 < b_2 < 1$. Which one of the following is NOT a necessary condition for forming a triangle using the three pieces?

Note: All lengths are in meter. The figure shown is representative.



(A) $b_1 < 0.5$ (B) $b_2 > 0.5$ (C) $b_2 < b_1 + 0.5$ (D) $b_1 + b_2 < 1$

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Q47. GATE ECE 2025

Open solution

The table lists the top 5 nations according to the number of gold medals won in a tournament; also included are the number of silver and the bronze medals won by them. Based only on the data provided in the table, which one of the following statements is INCORRECT?

Nation	Gold	Silver	Bronze
USA	40	44	41
Canada	39	27	24
Japan	20	12	13
Australia	17	19	16
France	16	26	22

(A) France will occupy the third place if the list were made on the basis of the total number of medals won.

(B) The order of the top two nations will not change even if the list is made on the basis of the total number of medals won.

(C) USA and Canada together have less than 50% of the medals awarded to the nations in the above table.

(D) Canada has won twice as many total medals as Japan.

Q48. GATE ECE 2025

Open solution

An organization allows its employees to work independently on consultancy projects but charges an overhead on the consulting fee. The overhead is 20\% of the consulting fee, if the fee is up to . $5,00,000$. For higher fees, the overhead is . $1,00,000$ plus 10% of the amount by which the fee exceeds . 5,00,000. The government charges a Goods and Services Tax of 18% on the total amount (the consulting fee plus the overhead). An employee of the organization charges this entire amount, i.e., the consulting fee, overhead, and tax, to the client. If the client cannot pay more than . 10,00,000, what is the maximum consulting fee that the employee can charge?

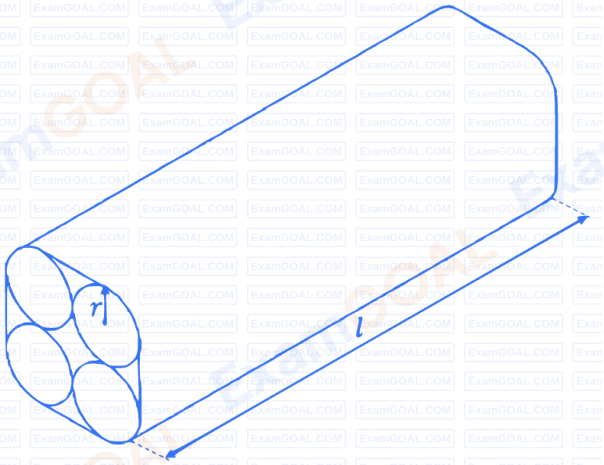
(A) $7,01,438$ (B) $7,24,961$ (C) $7,51,232$ (D) $7,75,784$

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Q49. GATE ECE 2024

Open solution

Four identical cylindrical chalk-sticks, each of radius $r = 0.5$ cm and length $l = 10$ cm, are bound tightly together using a duct tape as shown in the following figure.



The width of the duct tape is equal to the length of the chalk-stick. The area (in cm^2) of the duct tape required to wrap the bundle of chalk-sticks once, is

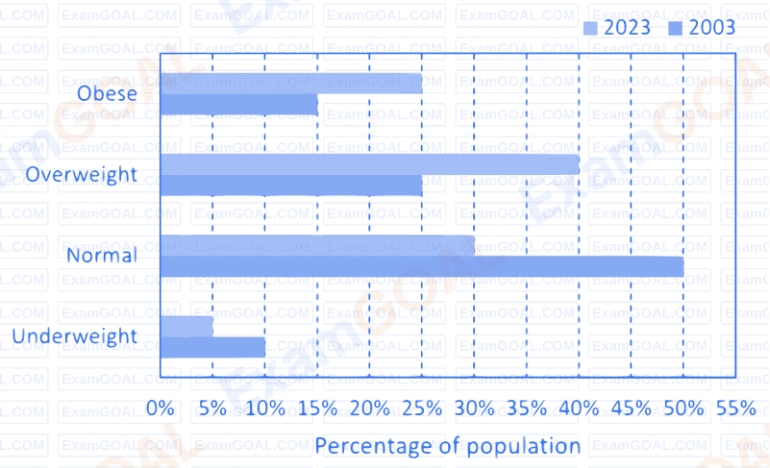
(A) 20 (4 + π) (B) 20 (8 + π) (C) 10 (8 + π) (D) 10 (4 + π)

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Q50. GATE ECE 2024

Open solution

The bar chart shows the data for the percentage of population falling into different categories based on Body Mass Index (BMI) in 2003 and 2023.



Based on the data provided, which one of the following options is INCORRECT?

(A) The ratio of the percentage of population falling into overweight category to the percentage of population falling into normal category has increased in 20 years.

(B) The ratio of the percentage of population falling into underweight category to the percentage of population falling into normal category has decreased in 20 years.

(C) The ratio of the percentage of population falling into obese category to the percentage of population falling into normal category has decreased in 20 years.

(D) The percentage of population falling into normal category has decreased in 20 years.

Q51. GATE ECE 2024

Open solution

Two identical sheets A and B, of dimensions 24 cm × 16 cm, can be folded into half using two distinct operations, FO1 or FO2.

In FO1, the axis of folding remains parallel to the initial long edge, and in FO2, the axis of folding remains parallel to the initial short edge.

If sheet A is folded twice using FO1, and sheet B is folded twice using FO2, the ratio of the perimeters of the final shapes of A and B is

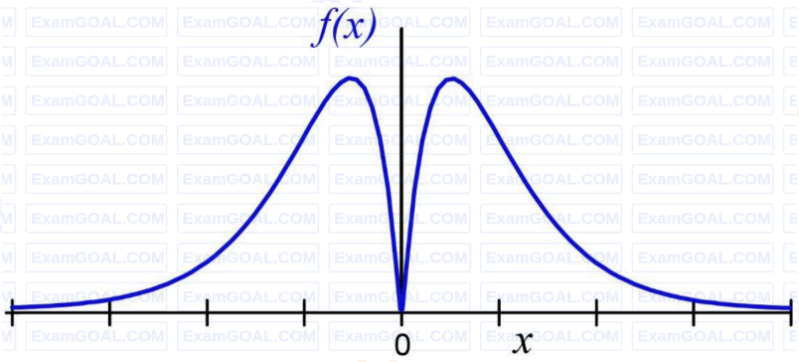
(A) 14:11 (B) 11:14 (C) 18:11 (D) 11:18

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Q52. GATE ECE 2023

Open solution

Which one of the following options represents the given graph?



(A) $f(x) = x^2 2^{-|x|}$ (B) $f(x) = x2^{-|x|}$ (C) $f(x) = |x|2^{-x}$ (D) $f(x) = x2^{-x}$

Q53. GATE ECE 2023

Open solution

Out of 1000 individuals in a town, 100 unidentified individuals are covid positive. Due to lack of adequate covid-testing kits, the health authorities of the town devised a strategy to identify these covid-positive individuals. The strategy is to:

(i) Collect saliva samples from all 1000 individuals and randomly group them into sets of 5.

(ii) Mix the samples within each set and test the mixed sample for covid.

(iii) If the test done in (ii) gives a negative result, then declare all the 5 individuals to be covid negative.

(iv) If the test done in (ii) gives a positive result, then all the 5 individuals are separately tested for covid.

Given this strategy, no more than _____ testing kits will be required to identify all the 100 covid positive individuals irrespective of how they are grouped.

(A) 700 (B) 600 (C) 800 (D) 1000

Q54. GATE ECE 2023

Open solution

A 100 cm $\times$ 32 cm rectangular sheet is folded 5 times. Each time the sheet is folded, the long edge aligns with its opposite side. Eventually, the folded sheet is a rectangle of dimensions 100 cm $\times$ 1 cm.

The total number of creases visible when the sheet is unfolded is _____.

(A) 32 (B) 5 (C) 31 (D) 63

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Q55. GATE ECE 2022

Open solution

Consider the following inequalities.

(i) $2x - 1 > 7$

(ii) $2x - 9 < 1$

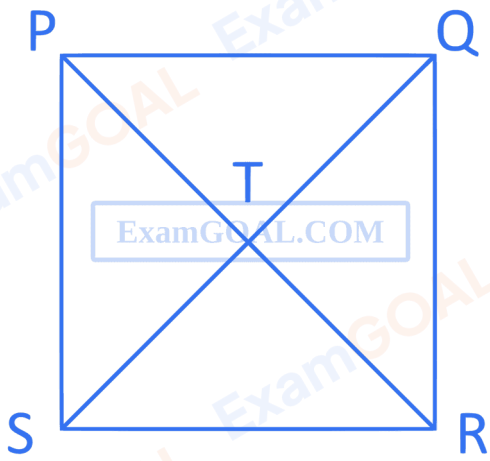
Which one of the following expressions below satisfies the above two inequalities?

(A) $x \leq -4$ (B) $-4 < x \leq 4$ (C) $4 < x < 5$ (D) $x \geq 5$

Q56. GATE ECE 2022

Open solution

Consider the following square with the four corners and the center marked as P, Q, R, S and T respectively.



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Let X, Y and Z represent the following operations :

X : rotation of the square by 180 degree with respect to the S-Q axis.

Y : rotation of the square by 180 degree with respect to the P-R axis.

Z : rotation of the square by 90 degree clockwise with respect to the axis perpendicular, going into the screen and passing through the point T.

Consider the following three distinct sequences of operation (which are applied in the left to right order).

(1) XYZZ

(2) XY

(3) ZZZZ

Which one of the following statements is correct as per the information provided above?

(A) The sequence of operations (1) and (2) are equivalent

(B) The sequence of operations (1) and (3) are equivalent

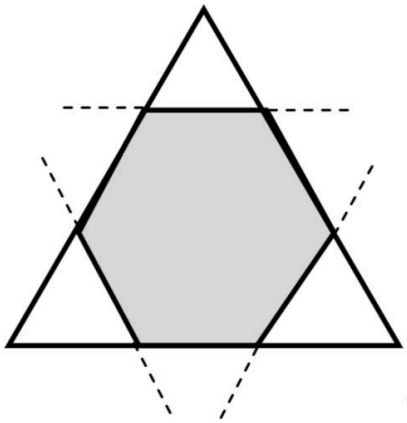
(C) The sequence of operations (2) and (3) are equivalent

(D) The sequence of operations (1), (2) and (3) are equivalent

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Q57. GATE ECE 2021

Open solution



Corners are cut from an equilateral triangle to produce a regular convex hexagon as shown in the figure above.

The ratio of the area of the regular convex to the area of the original equilateral triangle is

(A) 4 : 5 (B) 2 : 3 (C) 3 : 4 (D) 5 : 6

Q58. GATE ECE 2021

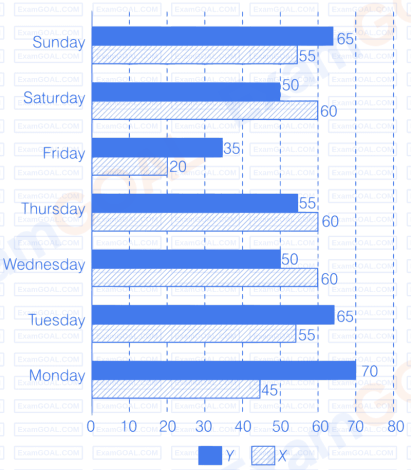
Open solution

Consider a square sheet of side 1 unit. In the first step, it is cut along the main diagonal to get two triangles. In the next step, one of the cut triangles is revolved about its short edge to form a solid cone. The volume of the resulting cone, in

(A) $\frac{2\pi}{3}$ (B) 3π (C) $\frac{\pi}{3}$ (D) $\frac{3\pi}{2}$

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Q59. GATE ECE 2021



Open solution

The number of minutes spent by two students, X and Y, excercising every day in a given week are shown in the bar chart above.

The number of days in the given week in which one of the students spent a minimum of 10% more than the other student, on a given day, is

(A) 7 (B) 5 (C) 4 (D) 6

Q60. GATE ECE 2020

a, b, c are real numbers. The quadratic equation $ax^2 - bx + c = 0$ has equal roots, which is β, then

(A) $\beta^3 = \frac{bc}{(2a^2)}$ (B) $\beta^2 = ac$

(C) $\beta = \frac{b}{a}$ (D) $b^2 \neq 4ac$

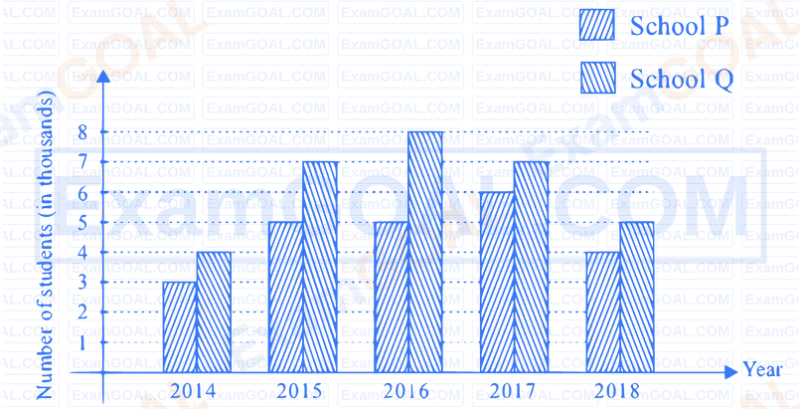
Open solution

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Q61. GATE ECE 2020

Open solution

The following figure shows the data of students enrolled in 5 years (2014 to 2018) for two schools P and Q. During this period, the ratio of the average number of the students enrolled in school P to the average of the difference of the number of students enrolled in schools P and Q is

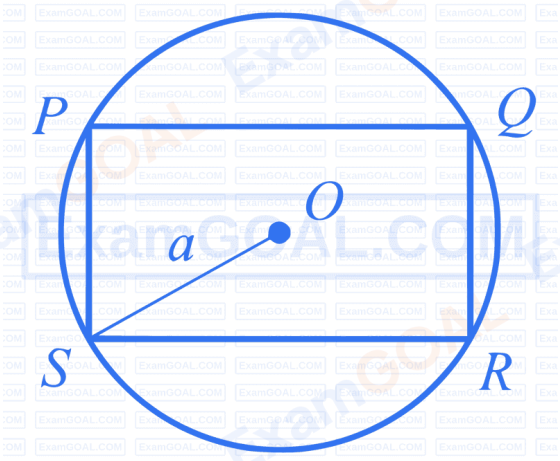


(A) $31 : 23$ (B) $23 : 8$ (C) $23 : 31$ (D) $8 : 23$

Q62. GATE ECE 2020

Open solution

A circle with center O is shown in the figure. A rectangle $PQRS$ of maximum possible area is inscribed in the circle. If the radius of the circle is a, then the area of shaded portion is ____ .



(A) $\pi a^2 - a^2$ (B) $\pi a^2 - 3a^2$ (C) $\pi a^2 - \sqrt{2}a^2$ (D) $\pi a^2 - 2a^2$

Q63. GATE ECE 2020

Open solution

It is quarter past three in your watch. The angle between the hour hand and the minute hand is ____ .

(A) 22.5° (B) 15^0 (C) 7.5^0 (D) 0^0

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Q64. GATE ECE 2019

Open solution

Two design consultants, P and Q, started working from 8 AM for a client. The client budgeted a total of USD 3000 for the consultants. P stopped working when hour hand moved by 210 degrees on the clock. Q stopped working when the hour hand moved by 240 degrees. P took two tea breaks of 15 minutes each during her shift, but took no lunch break. Q took only one lunch break for 20 minutes, but no tea breaks. The market rate for consultants is USD 200 per hour and breaks are not paid. After paying the consultants, the client shall have USD _____ remaining in the budget.

(A) 000.00 (B) 300.00 (C) 166.67 (D) 433.33

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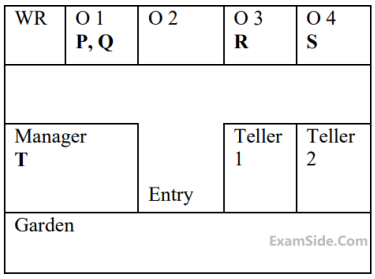
Q65. GATE ECE 2019

Five people P, Q, R, S and T work in a bank. P and Q don't like each other but have to share an office till T gets a promotion and moves to the big office next to the garden. R, who is currently sharing an office with T wants to move to the adjacent office with S, the handsome new intern.

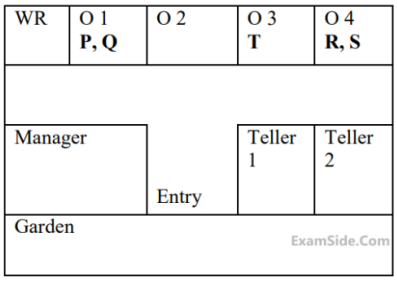
Open solution

Given the floor plan, what is the current location of Q, R and T ? [O = office, WR = Wash room]

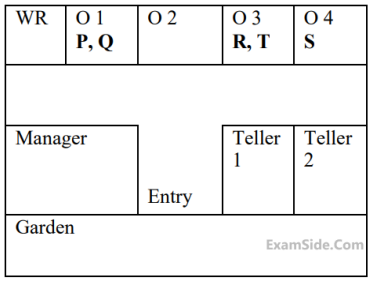
(A)



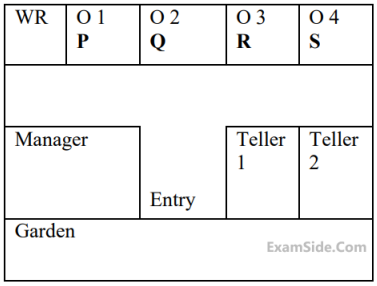
(B)



(C)



(D)

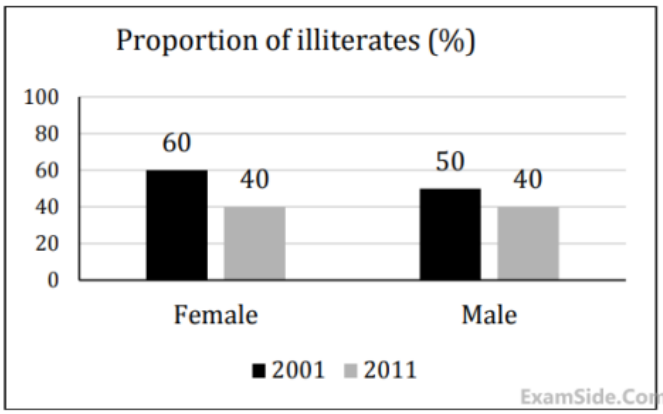


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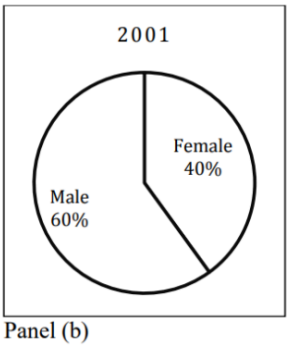
Q66. GATE ECE 2019

Open solution

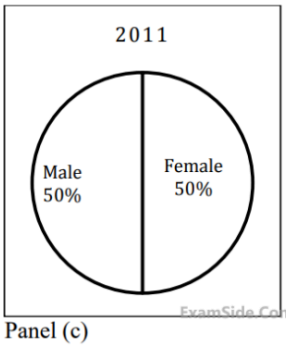
The bar graph in Panel (a) shows the proportion of male and female illiterates in 2001 and 2011. The proportions of males and females in 2001 and 2011 are given in Panel (b) and (c), respectively. The total population did not change during this period. The percentage increase in the total number of literates from 2001 to 2011 is _____.



Panel (a)



Panel (b)



Panel (c)

(A) 33.43 (B) 35.43 (C) 30.43 (D) 34.43

Q67. GATE ECE 2018

Open solution

Two alloys A and B contain gold and copper in the ratios of 2 : 3 and 3 : 7 by mass, respectively. Equal masses of alloys A and B are melted to make an alloy C. The ratio of gold to copper in alloy C is

(A) 6 : 11 (B) 5 : 10 (C) 7 : 13 (D) 9 : 13

Q68. GATE ECE 2018

Open solution

Leila aspires to buy a car worth Rs. 10,00,000 after 5 years. What is the minimum amount in Rupees that she should deposit now in a bank which offers 10% annual rate of interest, if the interest was compounded annually?

(A) 5,00,000 (B) 6,21,000 (C) 6,66,667 (D) 7,50,000

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Q69. GATE ECE 2018

A cab was involved in a hit and run accident at night. You are given the following data about the cabs in the city and the accident.

(i) 85% of cabs in the city are green and the remaining cabs are blue.

(ii) A witness identified the cab involved in the accident as blue.

(iii) It is known that a witness can correctly identify the cab colour only 80% of the time.

Which of the following options is closest to the probability that the accident was caused by a blue cab?

(A) 41% (B) 15% (C) 12% (D) 80%

Open solution

Q70. GATE ECE 2017 Set 2

Each of P, Q, R, S, W, X, Y and Z has been married at most once. X and Y married and have two children P and Q, Z is grandfather of the daughter 'S' of P. Further Z and W married and are parents of R. Which one of the following must necessarily be FALSE?

(A) X is the mother in law to R
(B) P and R are not married to each other
(C) P is son of X and Y
(D) Q cannot be married to R

Open solution

Q71. GATE ECE 2017 Set 2

The number of 3 digit numbers such that the digit 1 is never to the immediate right of 2 is

(A) 781 (B) 791 (C) 881 (D) 891

Open solution

Q72. GATE ECE 2017 Set 2

1200 men and 500 women can build a bridge in 2 weeks. 900 men and 250 women will take 3 weeks to build the same bridge. How many men will be needed to build the bridge in one week?

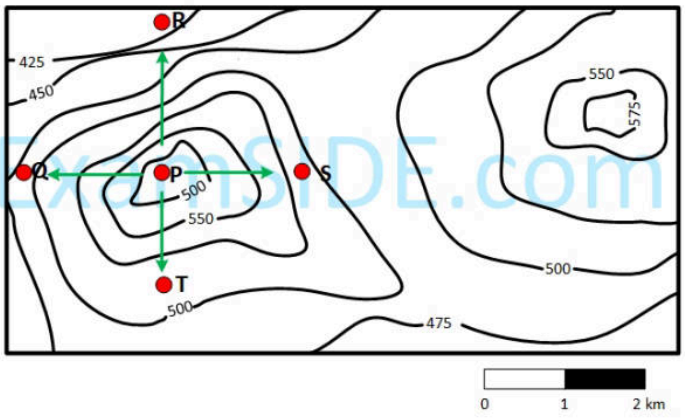
(A) 3000 (B) 3300 (C) 3600 (D) 3900

Open solution

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Q73. GATE ECE 2017 Set 2

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.



Open solution

Which of the following is the steepest path leaving from P?

(A) P to Q (B) P to R (C) P to S (D) P to T

Q74. GATE ECE 2017 Set 1

There are 3 Indians and 3 Chinese in a group of 6 people. How many subgroups of this group can we choose so that every subgroup has at least one Indian?

(A) 52 (B) 56 (C) 48 (D) 44

Open solution

Q75. GATE ECE 2017 Set 1

S, T, U, V, W, X, Y and Z are seated around a circular table. T's neighbors are Y and V. Z is seated third to the left of T and second to the right of S. U's neighbors are S and Y: and T and W are not seated opposite each other. Who is third to the left of V?

(A) X (B) W (C) U (D) T

Open solution

Q76. GATE ECE 2017 Set 1

Trucks (10 m long) and cars (5 m long) go on a single lane bridge. There must be a gap of at least 20 m after each truck and a gap of at least 15 m after each car. Trucks and cars travel at a speed of 36 km/h. If cars and trucks go alternately, what is the maximum number of vehicles that can use the bridge in one hour?

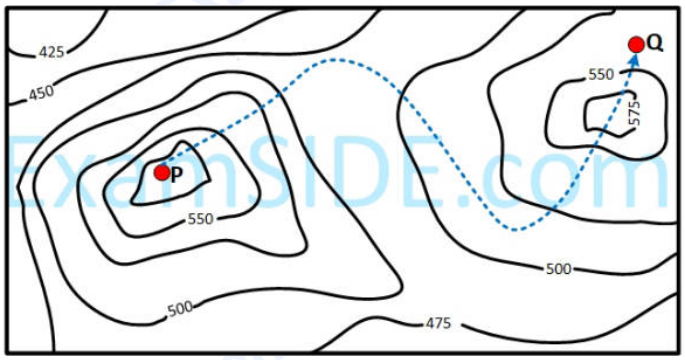
(A) 1440 (B) 1200 (C) 720 (D) 600

Open solution

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Q77. GATE ECE 2017 Set 1

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.



The path from P to Q is best described by

(A) Up-Down-Up-Down
(B) Down-Up-Down-Up
(C) Down-Up-Down
(D) Up-Down-Up

Open solution

Q78. GATE ECE 2014 Set 1

Find the odd one from the following group:
W,E,K,O
I,Q,W,A
F,N,T,X
N,V,B,D

(A) W,E,K,O
(B) I,Q,W,A
(C) F,N,T,X
(D) N,V,B,D

Open solution

Q79. GATE ECE 2014 Set 1

For submitting tax returns, all resident males with annual income below Rs 10 lakh should fill up Form P and all resident females with income below Rs 8 lakh should fill up Form Q. All people with incomes above Rs 10 lakh should fill up Form R, except non residents with income above Rs 15 lakhs, who should fill up Form S. All others should fill Form T. An example of a person who should fill Form T is

(A) a resident male with annual income Rs 9 lakh
(B) a resident female with annual income Rs 9 lakh
(C) a non-resident male with annual income Rs 16 lakh
(D) a non-resident female with annual income Rs 16 lakh

Open solution

Q80. GATE ECE 2014 Set 1

A train that is 280 meters long, traveling at a uniform speed, crosses a platform in 60 seconds and passes a man standing on the platform in 20 seconds. What is the length of the platform in meters?

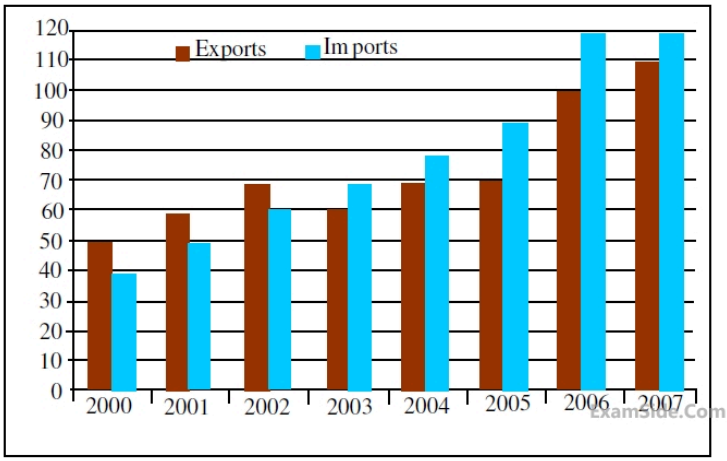
Open solution

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Q81. GATE ECE 2014 Set 1

The exports and imports (in crores of Rs.) of a country from 2000 to 2007 are given in the following bar chart. If the trade deficit is defined as excess of imports over exports, in which year is the trade deficit 1/5th of the exports?

Open solution



(A) 2005 (B) 2004 (C) 2007 (D) 2006

Q82. GATE ECE 2014 Set 1

You are given three coins: one has heads on both faces, the second has tails on both faces, and the third has a head on one face and a tail on the other. You choose a coin at random and toss it, and it comes up heads. The probability that the other face is tails is

Open solution

(A) 1/4 (B) 1/3 (C) 1/2 (D) 2/3

Q83. GATE ECE 2013

A car travels 8 km in the first quarter of an hour, 6 km in the second quarter and 16 km in the third quarter. The average speed of the car in km per hour over the entire journey is

Open solution

(A) 30 (B) 36 (C) 40 (D) 24

Q84. GATE ECE 2013

Find the sum to n terms of the series 10 + 84 + 734 +.....

Open solution

(A) $\frac{9(9^n+1)}{10}+1$ (B) $\frac{9(9^n-1)}{8}+1$
(C) $\frac{9(9^n-1)}{8}+n$ (D) $\frac{9(9^n-1)}{8}+n^2$

Q85. GATE ECE 2013

The set of values of p for which the roots of the equation $3x^2 + 2x + p(p - 1) = 0$ are of opposite sign is

Open solution

(A) $(-\infty, 0)$ (B) $(0, 1)$ (C) $(1, \infty)$ (D) $(0, \infty)$

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Q86. GATE ECE 2013

What is the chance that a leap year, selected at random, will contain 53 Sundays?

(A) 2/7 (B) 3/7 (C) 1/7 (D) 5/7

Open solution

Q87. GATE ECE 2012

The data given in the following table summarizes the monthly budget of an average household.

Category	Amount (Rs.)
Food	4000
Clothing	1200
Rent	2000
Savings	1500
Other expenses	1800

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Open solution

The approximate percentage of the monthly budget **NOT** spent on savings is

(A) 10 % (B) 14 % (C) 81 % (D) 86 %

Q88. GATE ECE 2012

A and B are friends. They decide to meet between 1 PM and 2 PM on a given day. There is a condition that whoever arrives first will not wait for the other for more than 15 minutes. The probability that they will meet on that day is

(A) 1/4 (B) 1/16 (C) 7/16 (D) 9/16

Open solution

Q89. GATE ECE 2012

Raju has 14 currency notes in his pocket consisting of only Rs. 20 notes and Rs. 10 notes. The total money value of the notes is Rs. 230. The number of Rs. 10 notes that Raju has is

(A) 5 (B) 6 (C) 9 (D) 10

Open solution

Q90. GATE ECE 2012

There are eight bags of rice looking alike, seven of which have equal weight and one is slightly heavier. The weighing balance is of unlimited capacity. Using this balance, the minimum number of weighings required to identify the heavier bag is

(A) 2 (B) 3 (C) 4 (D) 8

Open solution

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Q91. GATE ECE 2012

One of the legacies of the Roman legions was discipline. In the legions, military law prevailed and discipline was brutal. Discipline on the battlefield kept units obedient, intact and fighting, even when the odds and conditions were against them.

Which one of the following statements best sums up the meaning of the above passage?

(A) Thorough regimentation was the main reason for the efficiency of the Roman legions even in adverse circumstances.

(B) The legions were treated inhumanly as if the men were animals.

(C) Discipline was the armies' inheritance from their seniors.

(D) The harsh discipline to which the legions were subjected to led to the odds and conditions being against them.

Open solution

Q92. GATE ECE 2011

Three friends, R, S and T shared toffee from a bowl. R took 1/3rd of the toffees, but returned four to the bowl. S took 1/4rd of what was left but returned three toffees to the bowl. T took half of the remainder but returned two back into the bowl. If the bowl had 17 toffees left, how many toffees were originally there in the bowl?

(A) 38 (B) 31 (C) 48 (D) 41

Open solution

Q93. GATE ECE 2011

Given that f(y) = |y| / y, and q is any non-zero real number, the value of | f(q) - f(-q) | is

(A) 0 (B) -1 (C) 1 (D) 2

Open solution

Q94. GATE ECE 2011

The sum of n terms of the series 4 + 44 + 444 +.... is

(A) $(4/81)\left[10^{n+1} - 9n - 1\right]$ (B) $(4/81)\left[10^{n-1} - 9n - 1\right]$

(C) $(4/81)\left[10^{n+1} - 9n - 10\right]$ (D) $(4/81)\left[10^{n} - 9n - 10\right]$

Open solution

Q95. GATE ECE 2011

The horse has played a little known but very important role in the field of medicine. Horses were injected with toxins of diseases until their blood built up immunities. Then a serum was made from their blood. Serums to fight with diphtheria and tetanus were developed this way. It can be inferred from the passage that horses were

(A) given immunity to diseases (B) generally quite immune to diseases

(C) given medicines to fight toxins (D) given diphtheria and tetanus serums

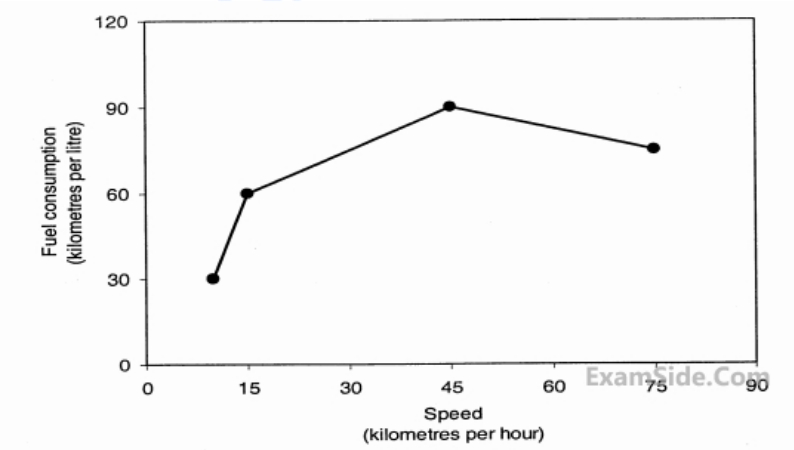
Open solution

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Q96. GATE ECE 2011

Open solution

The fuel consumed by a motorcycle during a journey while traveling at various speeds is indicated in the graph below.



The distances covered during four laps of the journey are listed in the table below

Lap	Distance (kilometres)	Average speed (kilometres per hour)
P	15	15
Q	75	45
R	40	75
S	10	10

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From the given data, we can conclude that the fuel consumed per kilometre was least during the lap

(A) P (B) Q (C) R (D) S

Q97. GATE ECE 2010

Open solution

Hari (H), Gita (G), Irfan (I) and Saira (S) are siblings (i.e. brothers and sisters). All were born on 1st January. The age difference between any two successive siblings (that is born one after another) is less than 3 years. Given the following facts:

i. Hari's age + Gita's age > Irfan's age + Saira's age.

ii. The age difference between Gita and Saira is 1 year. However, Gita is not the oldest and Saira is not the youngest.

iii. There are not twins.

In what order were they born (Oldest first)?

(A) HSIG (B) SGHI (C) IGSH (D) IHSG

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Q98. GATE ECE 2010

5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long will it take to build the wall?

(A) 20 days (B) 18 days (C) 16 days (D) 15 days

Open solution

Q99. GATE ECE 2010

Modern warfare has changed from large scale clashes of armies to suppression of civilian populations. Chemical agents that do their work silently appear to be suited to such warfare; and regretfully, there exist people in military establishments who think that chemical agents are useful tools for their cause.

Which of the following statements best sums up the meaning of the above passage:

(A) Modern warfare has resulted in civil strife.

(B) Chemical agents are useful in modern warfare.

(C) Use of chemical agents in warfare would be undesirable.

(D) People in military establishments like to use chemical agents in war.

Open solution

Q100. GATE ECE 2010

Given digits 2,2,3,3,3,4,4,4,4 how many distinct 4 digit numbers greater than 3000 can be formed?

(A) 50 (B) 51 (C) 52 (D) 54

Open solution

Q101. GATE ECE 2010

If 137+276=435 how much is 731+672?

(A) 534 (B) 1403 (C) 1623 (D) 1513

Open solution

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Answer Key

Marks 1

1. **B**	2. **B**	3. **B**	4. **A**	5. **B**
6. **C**	7. **A**	8. **C**	9. **C**	10. **C**
11. **D**	12. **B**	13. **B**	14. **C**	15. **D**
16. **D**	17. **B**	18. **A**	19. **A**	20. **B**
21. **A**	22. **C**	23. **B**	24. **A**	25. **D**
26. **D**	27. **B**	28. **B**	29. **C**	30. **B**
31. **A**	32. **C**	33. **C**	34. **A**	35. **A**
36. **C**	37. **B**	38. **A**	39. **C**	40. **C**
41. **D**	42. **A**	43. **D**		

Marks 2

44. **C**	45. **C**	46. **D**	47. **C**	48. **B**
49. **D**	50. **C**	51. **A**	52. **A**	53. **A**
54. **C**	55. **C**	56. **B**	57. **B**	58. **C**
59. **D**	60. **A**	61. **B**	62. **D**	63. **C**
64. **C**	65. **C**	66. **C**	67. **C**	68. **B**
69. **A**	70. **B**	71. **C**	72. **C**	73. **A**
74. **B**	75. **A**	76. **A**	77. **C**	78. **D**
79. **B**	80. **560**	81. **D**	82. **B**	83. **C**
84. **D**	85. **B**	86. **A**	87. **D**	88. **C**
89. **A**	90. **A**	91. **A**	92. **C**	93. **D**
94. **C**	95. **B**	96. **B**	97. **B**	98. **D**
99. **C**	100. **D**	101. **C**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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